The F4 tower engine over a generic prime field

Lane F, lead 4 · 2026-10-15 · branch `claude/f4-generic-field` · `research/f4_generic_field_20261015/`

Summary. `f4_tower_generic<F: PrimeField>` runs the PKM tower F4 engine of `f4_fp_tower` over any prime-field backend. At `Fp32` it reproduces `f4_fp_tower` field for field (182 rows, 0 differences) and is not slower (paired median ratio 1.09 and 1.06); above 2^{31} it is 2.18–2.24× faster; at 45 bits the new AVX-512 IFMA backend `Fp52` is 2.57× `Fp64`. All five pre-registered speed claims pass. With it, the Kummer tower runs at $v_2(p-1) = 40, 48, 56$ and at the NIST P-224 prime ($v_2 = 96$), and the Dickson torus tower at P-192 and P-256. In all 74 valued cells the solving degree D — and the width and step count — equal the recorded 31–32-bit values (H_same supported, none differing); on the fixed binary even the multiply-add count is identical across the five Kummer primes. Every finished row is verified (exhaustive decomposition count; points re-checked in `BigUint`). Host: 4 vCPU Xeon 2.10 GHz, shared with another lane (L0).

1. Engine and backends

- `f4_tower_generic`: the `u32` engine's pairs, criteria, symbolic preprocessing, Faugère–Lachartre elimination and full-rank exit, with coefficients `F::Elem` and dense kernels on Stage 0′s lazy accumulators (`row_load`/`row_sub`/`row_reduce`, sparse `row_scatter_axpy`); per-row product counting against `lazy_capacity`; block sizes independent of the backend, so the report is too.
- `Fp52` ($p < 2^{52}$): `Fp64`′s Montgomery elements; accumulator $\mathrm{lo} + \mathrm{hi} \cdot 2^{52}$ filled by `vpmadd52luq/huq`, 4 entries per 2 instructions; capacity 4095 products.
- `pkm_tower_pilot --engine generic --backend auto|…` (`Fp32` < 2^{32}, `Fp52` < 2^{52}, `Fp64` < 2^{63}, `FpP192`, `FpWide<3>`, `Fp224`, `FpWide<4>`), `--p P-192|P-224|P-256|proth:K`; above 2^{63} a builder generic over the backend (`pkm_tower_wide`) with `--verify`.

2. Correctness

gate	result
unit tests	7/7 + 7/7: every backend (`Fp32` vector/scalar, `Fp64`, `FpWide<1>`, `FpWide<4>`, `Fp224`, `Fp52` IFMA/scalar, capacity 1–3 wrappers) = `f4_fp_tower` report and basis on random rings at 5 primes; brute-force refutation and point counts at a 40-bit prime and P-224; backend agreement at 40, 48, 56 bits and P-224; `Fp52` vs `BigUint` through full capacity
G2 equality	`u32` vs generic `Fp32`: 42 + 28 + 56 + 56 rows (Kummer p_2, split Dickson p_1, torus T31, T24; tower and nulls; $m = 2, 3, 4$; $N \le 18$), **0 differences**; Kummer p_2 at `Fp64`, `Fp52`, `FpWide<4>`: 22 rows each, 0 differences
G3	arms of every speed cell agree on every field but the clock
G4	`pkm_tower_check verify`: 97/97 u64-builder rows agree; wide `--verify`: 82/82 rows ok (refutation ⟺ no decomposition; every decomposition and planted point satisfies every input equation and basis element in `BigUint`)

3. Speed (pre-registered)

7 interleaved repetitions, one thread, one core; F4 time per system.

id	claim	base ms	cand. ms	ratio	min–max	verdict
S1a	split Dickson p_1, $m = 2, N = 16$: `Fp32` vs `f4_fp_tower` (≥ 0.95)	3005	2791	1.091	0.900–1.123	PASS
S1b	torus T31, $m = 2, N = 16$: same (≥ 0.95)	2988	2784	1.058	0.969–1.096	PASS
S2a	Kummer $p_2 > 2^{31}$, $m = 2, N = 16$ (≥ 1.5)	8921	4009	2.175	2.034–2.306	PASS
S2b	Kummer p_2, $m = 3, N = 12$ (≥ 1.5)	15578	7001	2.240	2.114–2.270	PASS
S3	Kummer 45-bit, $m = 2, N = 16$: `Fp52` vs `Fp64` (≥ 1.8)	10333	4074	2.568	2.342–2.653	PASS

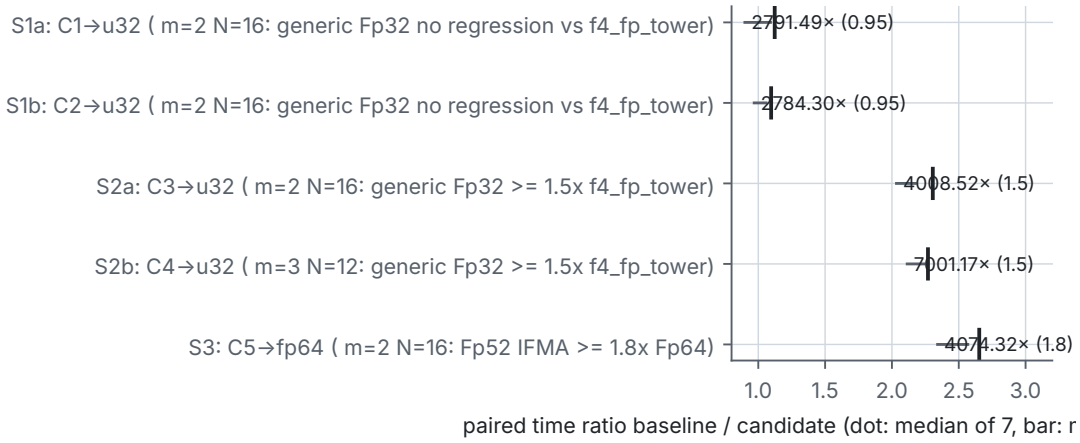


Figure 1: Paired ratios (dot: median, bar: min–max, tick: threshold).

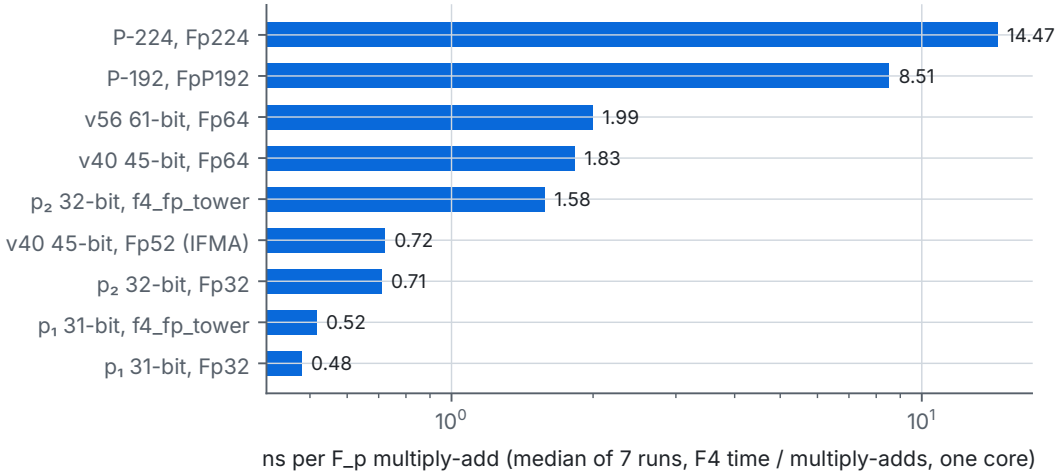


Figure 2: ns per F_p multiply-add in the engine: 0.48 (`Fp32`, 31-bit), 0.71 (`Fp32`, 32-bit; `f4_fp_tower` 1.58), 0.72 (`Fp52`, 45-bit; `Fp64` 1.83), 2.0 (`Fp64`, 61-bit), 8.5 (P-192), 14.5 (P-224) — 30× the 31-bit cost.

A 4-limb IFMA accumulator for P-224/P-256 was tried (exploratory) and dropped: correct, but 16.9 vs 12.4 ns per entry against `FpWide<4>` on this host.

4. Measured cells

Two random targets per cell; `D / width / steps`; reference: round 8 (Kummer $p_2 = 3 \cdot 2^{30} + 1$; torus T31) and round 2 (Kummer p_1, $m = 4$).

m	N	recorded	$v_2 = 40$	$v_2 = 48$	$v_2 = 56$	P-224
2	10	4/337/10	4/337/10	4/337/10	4/337/10	4/337/10
2	12	5/1231/13	5/1231/13	5/1231/13	5/1231/13	5/1231/13
2	14	5/2304/15	5/2304/15	5/2304/15	5/2304/15	5/2304/15
2	16	5/4789/17	5/4789/17	5/4789/17	5/4789/17	5/4789/17
2	18	5/8644/20	5/8644/20	5/8644/20	5/8644/20	5/8644/20
3	9	6/1186/16	6/1186/16	6/1186/16	6/1186/16	6/1186/16
3	12	6/5019/18	6/5019/18	6/5019/18	6/5019/18	6/5019/18
4	8	6/1049/21	6/1049/21	6/1049/21	6/1049/21	6/1049/21

$N = 2 \ldots 8$ ($m = 2$) and $N = 6$ ($m = 3$) also equal (README §4).

m	N	torus T31 (recorded)	P-192	P-256
2	12	5/1223/13	5/1223/13	5/1223/13
2	16	5/4786/17	5/4786/17	5/4786/17

2	18	5/8639/20	5/8639/20	—
3	9	6/1184/16	6/1184/16	6/1184/16
3	12	6/5018/18	6/5018/18	—

EXTENSION_TYP

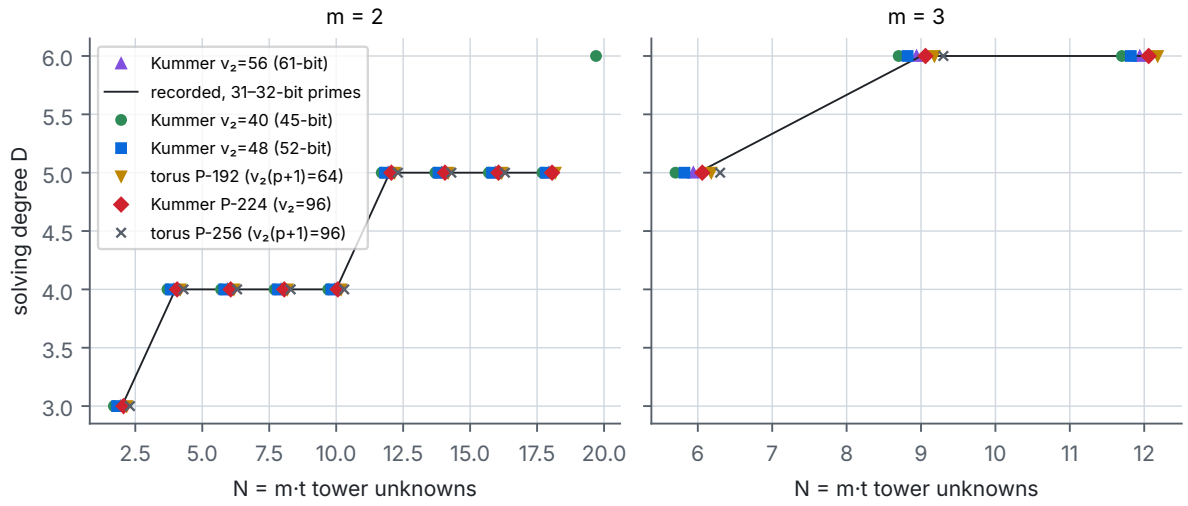


Figure 3: Solving degree against N; line: recorded small-prime D.

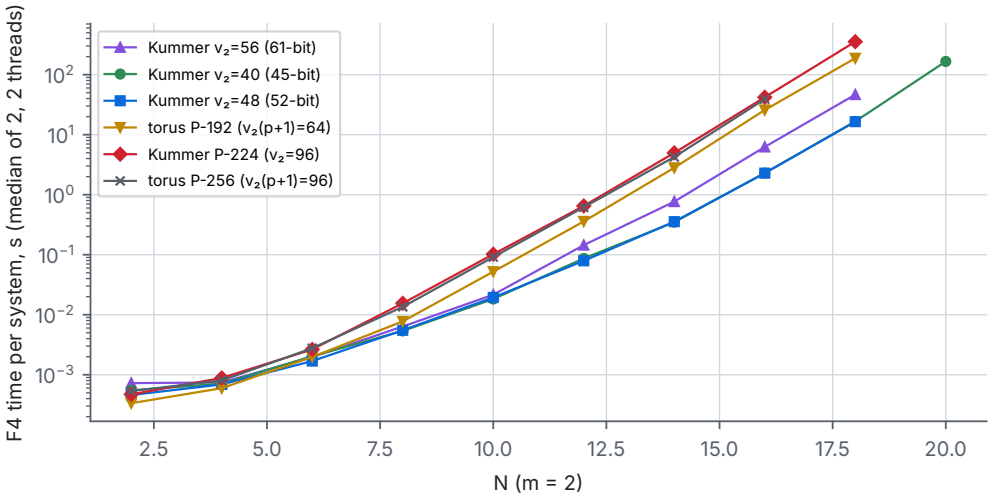


Figure 4: F4 time per system against N, $m = 2$ (two threads, shared host).

5. Reading and disclosures

H_same is supported in every series (PROTOCOL §6): no valued cell differs in D; width and steps agree too, and on the fixed binary so does the multiply-add count. The Kummer system's coefficients do not involve ζ, so $v_2(p-1)$ can enter only through accidental cancellation mod p — not seen.

Deviations: the first depth run sized S-row blocks by accumulator width, so multiply-adds depended on the backend; fixed and re-run (v1 vs v2: 0 differences in D, width, steps over 197 rows). The `FpWide<4>` equality arm on the old binary showed 4 multiply-add differences; re-run on the fixed binary, 0. A container restart interrupted the extension run, which was resumed. Conductor was unavailable.